

ARYAN KUMAR

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Education

IIT Bombay

B.Tech | Minor in Machine Learning and Data Science | **CPI: 8.23/10**

Oct 2022 – May 2026

Mumbai, India

Experience

Loop Health

June 2026 – Present

Software Engineer

Bangalore, India

- Designed a centralized **communications service** replacing per-service messaging code in **5** backends, modeling providers behind a channel abstraction with per-channel retries and webhook-driven delivery-status reconciliation, handling **200K+** monthly notifications over **SMS, WhatsApp, email** and **Slack**
- Instrumented **Datadog** APM with custom spans and monitors across **3** services, cutting detection of silent Slack and WhatsApp delivery failures from **hours to seconds** via provider-level error-rate alerting
- Architected **Blood Intel**, an event-driven biomarker pipeline on **Pub/Sub** processing **500** reports/day, with a typed event registry decoupling publishers from subscribers and **6** stages as **Temporal** workflows with idempotent retries
- Implemented a semantic retrieval layer over a **Neo4j** graph of **50K+** patient-biomarker relationships, resolving marker-condition paths into a constrained evidence set that bounds LLM output to retrieved facts
- Built **2** onboarding flows as **A/B** variants on a config-driven **state-machine** engine (**Spring Boot/jOOQ**) for **1,200** companies, driving transitions server-side and aggregating **10+** internal services into **SDUI** payloads
- Shipped the **20** corresponding **React Native** screens across iOS and Android, bridging native permission APIs and committing state on OS-dialog dismissal so users returning from Settings resume on the correct step
- Resolved **15+** production incidents on-call across distributed microservices, authoring RCAs adopted as team references

Mobavenue Media

Feb 2025 – Aug 2025

ML Intern | Founders Office

Mumbai, India

- Built and deployed a **TabNet** install/purchase propensity pipeline trained on **10B+** production ad impressions, reaching **85%** recall on installs and **70%** on purchases against heavily imbalanced conversion labels, lifting campaign ROAS
- Owned the core bidding engine end to end, fusing user propensity scores, inventory tiering and hourly pacing signals into one scoring path and cutting bid spend **18%** without loss in conversion volume
- Shipped a real-time scoring service for ad inventory, narrowing a **58K** candidate pool to the **8K** highest-value placements per request at sub-**4ms** p99 under live bidding traffic

Projects

uGit: Git From Scratch in Java | Self-Project | Github

Apr 2026 – May 2026

- Reimplemented **Git** plumbing in **dependency-free** Java - init, add, commit, branch, checkout, log - with ref resolution
- Built a **SHA-1** content-addressable store persisting **zlib**-deflated blobs, trees and commits as an immutable DAG
- Implemented a staging index diffing the working tree against **HEAD** by hash comparison, avoiding full-content reads

JPEG-like Image Compression Engine | Course Project, Guide: Prof. Ajit Rajwade | Github

Feb 2025 – Apr 2025

- Built a **JPEG** image compression pipeline from scratch using 8×8 **DCT**, quantization, and entropy coding
- Optimized quality factor for **80%** compression while limiting image degradation to under **5% PSNR** loss

Satellite Image Segmentation & Change Detection | ISRO, Smart India Hackathon

Nov 2023 – Dec 2023

- Ranked **5th** nationally among **200+** finalists, one of **3** IIT Bombay teams recognised by the Ministry of Education
- Developed a CNN based on the U-Net architecture to segment and detect temporal change, achieving accuracy of **92%**
- Extended the model into a scalable change-detection framework, proposed for ISRO's monitoring pipeline across borders

Technical Skills

Languages: C/C++, Java, Python, SQL, JavaScript/TypeScript

Observability & Tools: Datadog, Sentry, LangSmith, Postman, DBeaver

Databases & Caching: PostgreSQL, MySQL, MongoDB, Neo4j, Redis, SQLite

Cloud & Infra: GCP, AWS, Kubernetes, Docker, Terraform, Linux, CI/CD, Kafka

AI/ML: RAG, Knowledge graphs, Agentic pipelines, LangChain, HuggingFace, PyTorch

Coursework: DSA, Operating Systems, Computer Networks, DBMS, OOP, System Design, Discrete Mathematics

Backend & Distributed Systems: Spring Boot, Node.js, REST, GraphQL, WebSockets, Pub/Sub, Temporal.io